

# Supplementary Material:

## A reduced viscoelastic FDTD formulation for ultrasound-driven shear wave propagation in soft tissue

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### S1 Theta-Burst TUS Protocol

The pipeline supports a theta-burst transcranial ultrasound stimulation (TUS) protocol motivated by neuromodulation applications.

**Protocol parameters.** The theta-burst envelope consists of 360  $\mu\text{s}$  tone-bursts repeated at a 1 kHz pulse repetition frequency (PRF), grouped into trains of 3 bursts. Each 3-burst group occupies the first 3 ms of a 200 ms theta period (1/5 Hz), giving an overall duty cycle of  $\sim 0.5\%$ . One complete theta cycle (200 ms) is simulated.

**Intensity scaling.** The acoustic simulation was performed at a source pressure of 1.5 MPa, yielding a measured focal  $I_{\text{SPPA}} \approx 633 \text{ W/cm}^2$  (after CW envelope correction). Typical neuromodulation protocols operate at lower intensities,  $I_{\text{SPPA}} \approx 20\text{--}30 \text{ W/cm}^2$ . Rather than re-running the acoustic simulation, the body force is scaled by the ratio of target-to-simulated focal intensity:

$$\mathbf{f}_{\text{neuro}} = \frac{I_{\text{SPPA}}^{\text{target}}}{I_{\text{SPPA}}^{\text{sim}}} \mathbf{f}_{\text{sim}}, \quad (\text{S1})$$

since the radiation force is proportional to acoustic intensity. For a target of  $30 \text{ W/cm}^2$  the scaling factor is  $30/633 \approx 0.047$ , reducing the peak axial force from  $\sim 41,800$  to  $\sim 2,000 \text{ N/m}^3$ .

**Results.** With the scaled ARF and theta-burst envelope, the shear FDTD solver (12,035 time steps over 200 ms) produces a peak displacement of  $|u_x|^{\text{max}} \approx 0.3 \text{ nm}$  at the end of the theta cycle. Instantaneous displacement during the 3 ms burst window reaches  $\sim 0.08 \mu\text{m}$  before viscous damping ( $\eta = 0.5 \text{ Pa}\cdot\text{s}$ ) dissipates the motion during the subsequent 197 ms off-period. The sub-nanometer residual at cycle end is consistent with the very low duty cycle (0.5%).

### S2 From Tissue Displacement to Membrane Tension

In neuromodulation applications, the quantity that ultimately gates mechanosensitive ion channels is not tissue displacement itself but the *membrane tension*  $T$  (force per unit length, mN/m) experienced by the lipid bilayer in which the channel is embedded.

The linearized strain tensor  $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$  is computed from the solver displacement field. Because the tissue is nearly incompressible, the mechanically relevant component is the deviatoric strain norm  $\varepsilon_{\text{dev}}$ . The effective membrane tension is then

$$T = K_A \alpha \varepsilon_{\text{dev}}, \quad (\text{S2})$$

where  $K_A$  is the area-expansion modulus of the lipid bilayer (typically  $0.1\text{--}0.5 \text{ N/m}$  for biological membranes) and  $\alpha$  is a dimensionless coupling factor relating far-field tissue strain to local membrane area change.

The tension  $T$  enters the Boltzmann gating model for mechanosensitive channels. Each channel opens with a steady-state probability  $P_o(T) = [1 + \exp(-\Delta A (T - T_{1/2})/k_B T_{\text{abs}})]^{-1}$ , where  $\Delta A$  is the effective gate area change and  $T_{1/2}$  is the half-activation tension.

For the channels most relevant to ultrasonic neuromodulation, the best-characterized half-activation tensions are: Piezo1,  $T_{1/2} = 2.7 \pm 0.4$  mN/m; and TRAAK,  $T_{1/2} \approx 5$  mN/m. Values for TREK-1 ( $\sim 3.5$  mN/m) and TREK-2 ( $\sim 4$  mN/m) carry larger uncertainty.

These tension scales set the target for the simulation pipeline: the shear wave solver must resolve displacement gradients accurately enough that the resulting membrane tensions can be compared with the channel  $T_{1/2}$  values. For a representative  $K_A = 0.25$  N/m and  $\alpha = 1$ , a deviatoric strain of  $\sim 10^{-5}$  produces a membrane tension of  $\sim 2.5$   $\mu$ N/m—well below channel thresholds—while strains of order  $10^{-2}$  reach the few-mN/m range where channel gating becomes significant.

## References